Vihaan Manchanda

Duke University · Explainable AI

prompting-explained

Which Words in Your Prompt Actually Matter?

A teaching tool that shows you the impact of each phrase in your prompt on the AI's answer.

SCAN TO TRY THE APP

github.com/Vihaan8/prompting-explained

01 Executive summary

People in almost every field now use AI tools, but there is no tool that teaches prompt-writing literacy. AI developers learn what works through practice. Most users never do. In one study, non-experts wrote prompts the way they would instruct a person, and struggled to fix them when the answer went wrong (Zamfirescu-Pereira et al., 2023). Small wording changes matter too. Changing only a prompt's formatting moved one model's accuracy by up to 76 points (Sclar et al., 2024). A badly worded prompt wastes tokens and context, and gets a worse answer.

I built **prompting-explained** to teach how each word in a prompt contributed to the AI's answer. It removes one phrase at a time, reruns the prompt, and highlights the phrases that changed the result. On 240 test prompts across 8 everyday tasks, the phrases it ranked highest mattered clearly more than random ones, and its top pick repeated 78% of the time.

The goal is a training tool companies load with their own tasks, so staff learn what works on their real work.

02 The problem

People learn to prompt by trial and error. Chat tools return an answer but never show which words the model acted on, so nobody can tell which of their habits help. For a company rolling out AI, that costs three things:

- **Worse answers.** Missing or counterproductive instructions go unnoticed.
- **Wasted tokens.** In my 240 test prompts, 45% of phrases asked for something the AI would have done anyway.
- **Hidden risk.** An answer can look fine while quietly breaking an instruction, such as keeping private details out, and nothing warns the user.

Prompting guides offer general tips. They don't show people what their own prompt did.

03 Who it's for

Everyday users. People who just want the right answer. They see which of their words helped and which didn't, without needing to know how AI works.

Employees at companies adopting AI. Staff who are still learning to use AI practice on their own company's tasks, documents and rules.

Governance and compliance teams. Test whether a safety instruction, such as "leave out customer details", actually changes the answer before rolling it out.

THE QUESTION

Can removing one phrase at a time and rerunning the prompt show which words changed the AI's answer, in a way that is **faithful** and **stable**?

04 Methods

1 Split the prompt into phrases

An AI splits the prompt into up to 10 phrases and tags each one (task, audience, format, politeness…). These phrases are the pieces the tool will perturb*. Code keeps each phrase exactly as typed, typos included.

*** Perturbation:** changing part of an input to see how the output responds. LIME does this with single words.

2 Remove one phrase and rerun

The full prompt runs 3 times. Then each phrase is deleted in turn and the shorter prompt runs twice. This is occlusion*. A phrase's effect is how much the score drops without it, which gives every phrase its own counterfactual**. Phrases that change nothing alone are also removed together once, in case they cover for each other.

*** Occlusion (leave-one-out):** hide one part of the input, rerun, and measure the change.
**** Counterfactual:** the same case with one thing changed, to see what that one thing did.

3 Score every answer two ways

Against the task. Each task comes with a short checklist that code grades automatically, like a teacher's rubric, from 0 (misses everything) to 1 (passes everything). An answer that doesn't do the task at all, such as asking "what should I explain?", scores 0.

Against your instructions. A second AI reads each answer and marks which of your instructions it followed*, such as "use a candy example", which code can't check. That separates "the AI followed this" from "this made the answer better".

*** AI as a judge:** using one AI to grade another's output. It can be wrong, so its mistakes are reported.

4 Filter out noise

The AI's answers vary from run to run, so every version runs more than once. A change only counts if it's bigger than the noise floor*. Anything smaller stays grey, and the tool says "not clear" rather than guessing.

*** Noise floor:** the smallest change the tool can tell apart from the AI's natural randomness (two standard errors).

5 Explain it in plain words

Each phrase that mattered gets one sentence, written by an AI, describing how the answer changed without it. Every number and every color comes from code.

05 Worked example

Task: write a prompt that gets the AI to explain inflation to your 10-year-old cousin. One user wrote this:

> Hi there! Explain inflation to my 10-year-old cousin. Use exactly 3 bullet points. Mention a candy bar in one of them. Use proper technical terms.

Made the answer better · Followed, no change in score · Worked against the task · No real difference

Green · "Explain inflation": without it, the AI explained how the body uses food for energy.

Blue · "Mention a candy bar": the AI did it, but the score was the same either way.

Red · "Use proper technical terms": the AI used words like "Consumer Price Index", too hard for a 10-year-old. Without it, the score rose from 0.40 to 0.78.

Grey · "Hi there!": a greeting gives the AI nothing to act on.

Scores run 0 to 1. On a fresh rerun, 5 of 6 verdicts repeated. The "without it" sentences are written by an AI and labelled as descriptions; every number is computed by code.

06 How it helps

How the tool addresses each problem from section 02, with what it showed on my 240 test prompts.

Worse answers

It shows which phrases drive the answer and which work against it. In the worked example, removing "Use proper technical terms" raised the score from 0.40 to 0.78. Across all the test prompts, format instructions and the task itself moved the score most (chart, right).

Wasted tokens

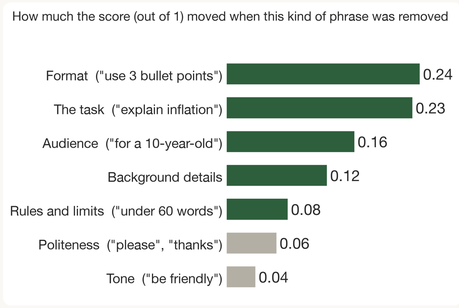
Phrases that change nothing show up grey, so people can see what to drop. Politeness and tone moved the score by only 0.06 and 0.04 on average.

Hidden risk

A second check reports whether each instruction was actually followed, and phrases that backfire show up red. In one prompt, "dont guess" made the AI speculate *more*, in all 5 runs.

How much each kind of phrase moved the score

Average change in score when that kind of phrase was removed, across 741 phrases.



07 Can the tool be trusted?

A tool that points at the wrong phrases would teach the wrong habits, so I tested it. Removing the two phrases it ranked highest dropped the score by **0.30**, against **0.18** for two random phrases (79 prompts, p = 0.0002), so it is faithful. Rerunning the whole analysis picked the same top phrase **78%** of the time, so it is stable. It falls short in three places:

- When an attached document already holds the answer, removing two phrases moved the score by less than 0.07, so there is little to rank.
- The AI that checks instructions can't count words. It passed 93 answers that broke a word limit.
- More reruns barely help. Nearly tripling the calls raised agreement from 83% to 91%.

08 Business implications

- **Faster adoption.** Employees learn what works on their own company's tasks, documents and rules, instead of from general tips.
- **Safer rollout.** Compliance teams can check that a safety instruction actually changes the answer before it goes into shared prompts.
- **Lower cost (stretch goal).** Once enough phrases are measured, results can be cached and a small local model trained. An app or browser extension could then flag wasted words before a prompt is sent, without an extra AI call.

09 Limitations and next steps

- One model (Claude Haiku 5.5), on 8 tasks with checks written in code.
- The 240 test prompts were written by an AI imitating users; I wrote 32 more messy ones. There is no study with real users yet.
- Each phrase is compared only with deleting it, not with a better phrase.
- Phrases that do nothing alone are retested as one group, but not in every combination. Shapley values* would, at a much higher cost.
- **Next:** a study with real users, more models, and Shapley-style scoring.

*** Shapley values:** give each part a fair share of the credit by testing it in every combination of parts.

REFERENCES Zamfirescu-Pereira et al. (2023), Why Johnny Can't Prompt, CHI · Sclar et al. (2024), prompt formatting sensitivity, ICLR · Zeiler & Fergus (2014), occlusion · Ribeiro et al. (2016), LIME · Lundberg & Lee (2017), SHAP · Petsiuk et al. (2018), deletion test · DeYoung et al. (2020), faithfulness · Alvarez-Melis & Jaakkola (2018), stability · Turpin et al. (2023), unfaithful model explanations